

Q2 Series Q

Hands-on I: the standard pipeline on one sample from matrix to cell types

10x's public GBM 5k, 5,604 cells. From loading to named clusters and state scores for malignant cells.

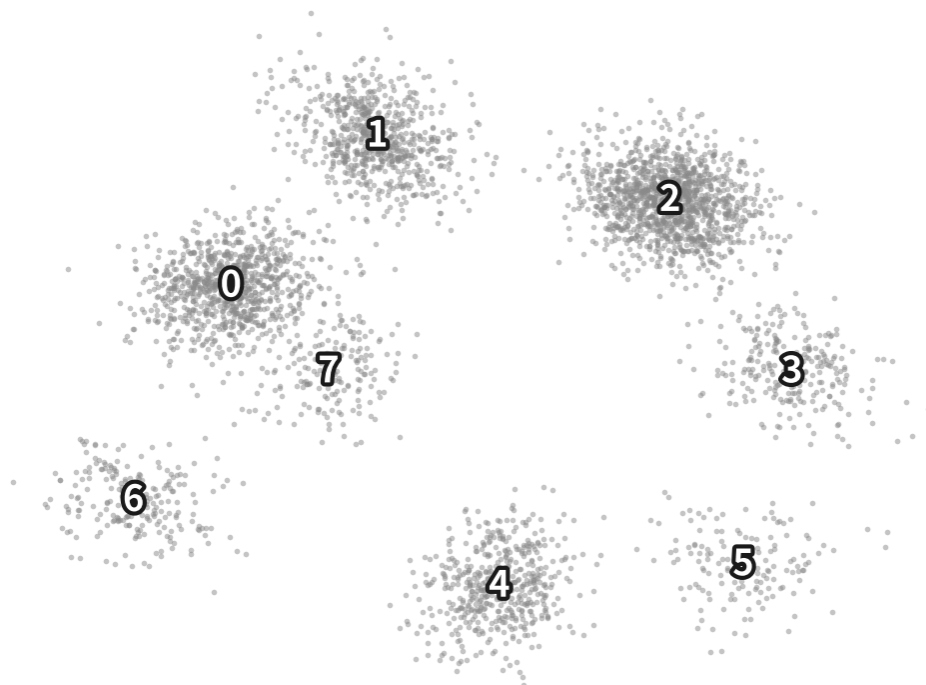
About 55 min | Prerequisites: Q1 | Scripts: R/00–03 | Deeper: B1, B5–B10

Fifty-five minutes from now you will have this

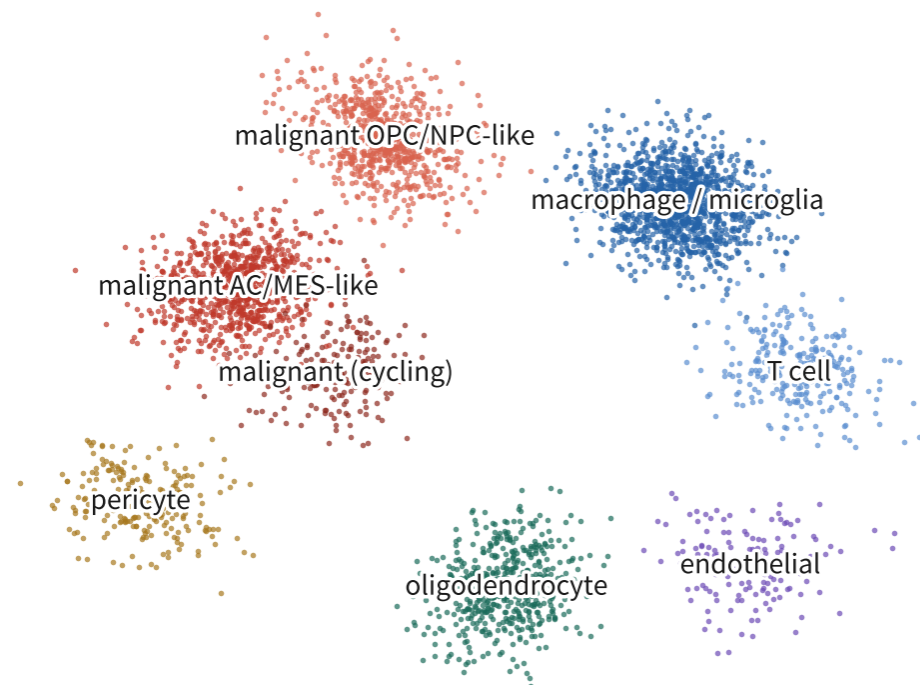
This episode's endpoint: from numbers to names

schematic / simulated

after clustering: eight numbers



after annotation: eight names



Numbers are arbitrary (sorted by size) and reshuffle with any parameter; names are what goes into the paper

Eight clusters, each named; malignant cells labelled 'glial / pending CNV' rather than declared tumour — and every parameter defensible

THE RULE FOR THIS EPISODE

**Every code block answers two questions:
what this line does, and why the parameter is set that
way.**

Full code lives in three practice scripts (R/01–03), each ending with exercises. 'Deeper: Bx' points to the Series B episode that goes all the way.

What you will do

- 1 Load GBM 5k into Seurat v5, set three QC cutoffs from this dataset's distributions, flag doublets
- 2 Run normalization → HVG → scaling → PCA → clustering → UMAP with three sanity checks (PC1, nPC, per-cluster QC)
- 3 Draw the four continents with gate genes, name clusters with markers + SingleR, score malignant states with Neftel gene sets

01

Environment and data

Script 00: install, download, set up the project

Practice data: a real tumour sample (GBM 5k)

Practice data: Human Glioblastoma Multiforme 5k

source	10x Genomics public dataset (Chromium 3' v3)
tissue	dissociated human glioblastoma tumour
cells	5,604 (called by Cell Ranger 4.0.0)
files	filtered_feature_bc_matrix.tar.gz (≈30 MB)
why	A real tumour: malignant, immune, vascular and normal brain cells together; runs on a laptop in 5 min

What it cannot answer

- ✗ One patient only: no patient-level inference (n=1)
- ✗ No matched normal: malignant calling relies on internal references (immune, oligodendrocytes)
- ✗ No clinical data: cannot link to response or survival
- Q3 switches to a 4-patient dataset to fill these gaps

Read the right side first: what a dataset can and cannot answer matters more than its cell count

Install, pin the version, set the seed

```
install.packages(c("Seurat", "dplyr", "ggplot2", "clustree"))  
BiocManager::install(c("scDb1Finder", "SingleR", "cellidex"))  
library(Seurat); library(dplyr); library(ggplot2)  
stopifnot(packageVersion("Seurat") >= "5.0.0") # v4 教學碼在 v5 會炸  
set.seed(1234) # 全系列固定  
for (d in c("data", "R", "output")) dir.create(d, showWarnings = FALSE)
```

No setwd(), no absolute paths: open an RStudio Project and the root is the working directory | Script 00 | Deeper: B1

Download, extract, load

```
url <- paste0("https://cf.10xgenomics.com/samples/cell-exp/4.0.0/",  
             "Parent_SC3v3_Human_Glioblastoma/",  
             "Parent_SC3v3_Human_Glioblastoma_filtered_",  
             "feature_bc_matrix.tar.gz")  
download.file(url, "data/gbm5k.tar.gz", mode = "wb") # 約 30 MB  
untar("data/gbm5k.tar.gz", exdir = "data/gbm5k")  
  
mtx.dir <- "data/gbm5k/filtered_feature_bc_matrix"  
gbm.data <- Read10X(data.dir = mtx.dir) # 吃資料夾  
gbm <- CreateSeuratObject(counts = gbm.data, project = "gbm5k",  
                          min.cells = 3, min.features = 200)
```

gbm

An object of class Seurat
~25,000 features across ~5,500 samples within 1 assay

min.cells / min.features are a coarse screen; real cutoffs come from distributions next. Use the numbers you actually get | Script 01 §1

First touch: dimensions, sparsity, gene naming

```
dim(gbm) # 基因 × 細胞
cnt <- LayerData(gbm, layer = "counts")
1 - Matrix::nnzero(cnt) / prod(dim(cnt)) # 稀疏度 : > 0.9 才正常
summary(Matrix::colSums(cnt)) # 每顆細胞的 UMI 總數
grep("^MT-", rownames(gbm), value = TRUE) # 人 MT-、鼠 mt-
head(grep("^RP[LS]", rownames(gbm), value = TRUE)) # 核糖體
cnt[c("PTPRC", "SOX2", "MBP"), 1:5] # 三個門牌基因
```

```
[1] 25xxx 55xx
[1] 0.93
  Min. 1st Qu. Median   Mean 3rd Qu.  Max.
  5xx   2xxx   4xxx   6xxx   8xxx  6xxxx
[1] "MT-ND1" "MT-ND2" "MT-CO1" ... (13 genes)
```

Thirty seconds that prevent three accidents: wrong species (MT- vs mt-), loading the raw matrix (hundreds of thousands of columns), Ensembl IDs instead of symbols | Script 01 §1

raw, filtered, and the 'soup'

Cell Ranger hands you two matrices; we load filtered, but raw has one more use

raw matrix



`raw_feature_bc_matrix`: every barcode (hundreds of thousands), mostly empty droplets. The RNA in empty droplets is a sample of the background 'soup'

filtered matrix



`filtered_feature_bc_matrix`: barcodes Cell Ranger calls cells via EmptyDrops (5,604 here). This is what we load; doublets and damaged cells are still inside

Ambient RNA



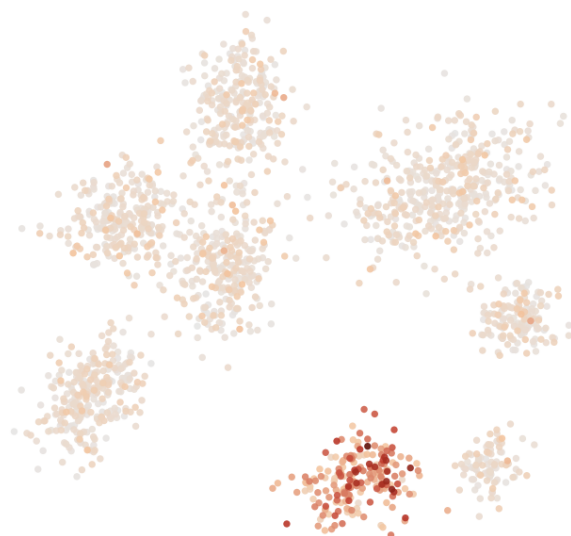
mRNA from lysed cells drifts into every droplet: each cell carries a little of everyone else. Worst in necrotic, slowly dissociated tumours

Ambient RNA: how to spot it, how to remove it

Ambient RNA: every cell in the matrix carries a little 'soup'

schematic / simulated

before: MBP



every cell carries a little MBP
—not because oligodendrocytes
are everywhere, but because
mRNA from lysed cells drifts
into every droplet

after SoupX: MBP



background estimated and
subtracted; the marker
lights only where it should

What SoupX does

- ① estimate the soup profile from empty droplets (low-UMI barcodes in the raw matrix)
- ② estimate each cell's contamination fraction p from genes a cell type cannot express (e.g. MBP outside oligodendrocytes)
- ③ subtract $p \times \text{soup}$ from every cell, return corrected integer counts

When it is mandatory: necrotic, slowly dissociated tumours; $p > 10\%$; when gate genes (PTPRC, MBP, HBB) show a little everywhere

A gate gene showing a little everywhere $\rightarrow$ estimate contamination first; above $p \approx 10\%$, run SoupX (or CellBender) before QC

SoupX: estimate and subtract the soup (optional, script 01 §0)

```
library(SoupX)
sc <- load10X("data/gbm5k_outs/")           # 需 raw + filtered 同在
sc <- setClusters(sc, gbm$seurat_clusters)    # 先粗分群
sc <- autoEstCont(sc)                        # 估污染比例 rho
sc$fit$rhoEst                               # 例: 0.06 → 可做可不做
adj <- adjustCounts(sc, roundToInt = TRUE)    # 校正後整數 counts
gbm <- CreateSeuratObject(adj, project = "gbm5k",
                           min.cells = 3, min.features = 200)
```

Needs the raw matrix, so datasets that ship filtered-only cannot do this; always keep raw for your own data. Skip below $p \approx 5\%$ | Deeper: B5

02

QC: three cutoffs that follow the data

Look first, cut second; save the raw object before filtering

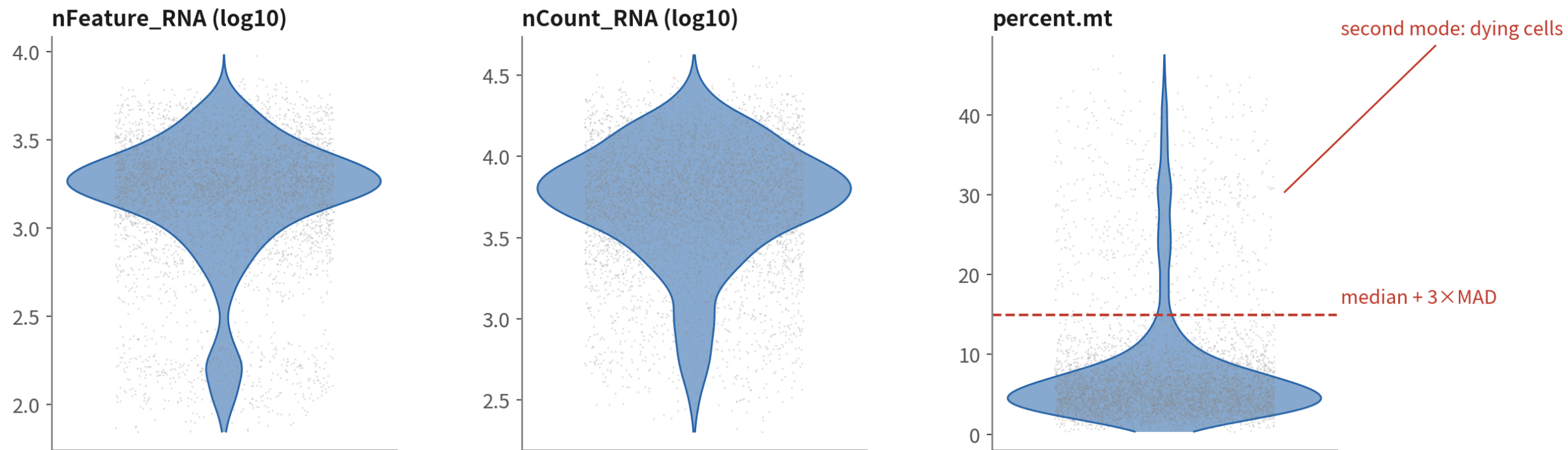
Three metrics: look before you cut

```
gbm[["percent.mt"]] <- PercentageFeatureSet(gbm, pattern = "^MT-")  
saveRDS(gbm, "output/gbm_raw.rds")      # 過濾前先存：截斷之後回不去  
  
VlnPlot(gbm, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"),  
        ncol = 3, pt.size = 0.1)  
FeatureScatter(gbm, "nCount_RNA", "percent.mt")  
FeatureScatter(gbm, "nCount_RNA", "nFeature_RNA")
```

nCount catches empties and doublets, nFeature debris, percent.mt dying cells | Script 01 §2 | Deeper: B5

QC distributions: a tumour sample as the example

The three QC metrics (typical tumour sample)

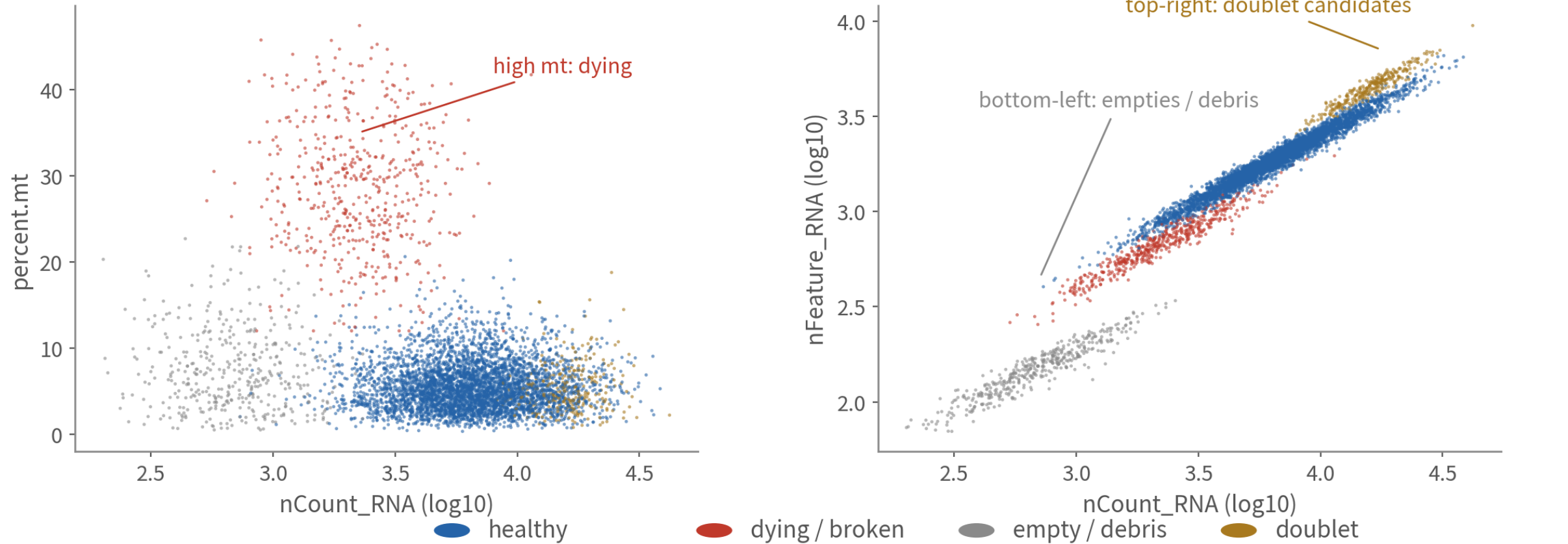


Tumour percent.mt runs higher than PBMC (malignant cells are metabolically active) and is often bimodal. A copied 5% cut removes a large slice of live cells

percent.mt runs higher than PBMC and is often bimodal. The second mode is dying cells; the line goes between the modes

Two scatters, each kind of junk in its own corner

Two scatters read together: each kind of junk owns a corner



One metric misleads; read them in pairs. The top-right doublet candidates get confirmed with a dedicated tool shortly

Three MAD cutoffs, then filter

```
mad.hi <- function(x, k = 3) median(x) + k * mad(x)    # mad() 已含 1.4826
mad.lo <- function(x, k = 3) median(x) - k * mad(x)

mt.hi  <- mad.hi(gbm$percent.mt)                      # 例 : ~ 12.8 ( 依你的資料 )
nf.lo  <- 10^mad.lo(log10(gbm$nFeature_RNA))          # 在 log 尺度算下限
nc.hi  <- 10^mad.hi(log10(gbm$nCount_RNA))            # 在 log 尺度算上限
c(mt.hi = mt.hi, nf.lo = nf.lo, nc.hi = nc.hi)

gbm <- subset(gbm, subset = percent.mt < mt.hi &
              nFeature_RNA > nf.lo & nCount_RNA < nc.hi)

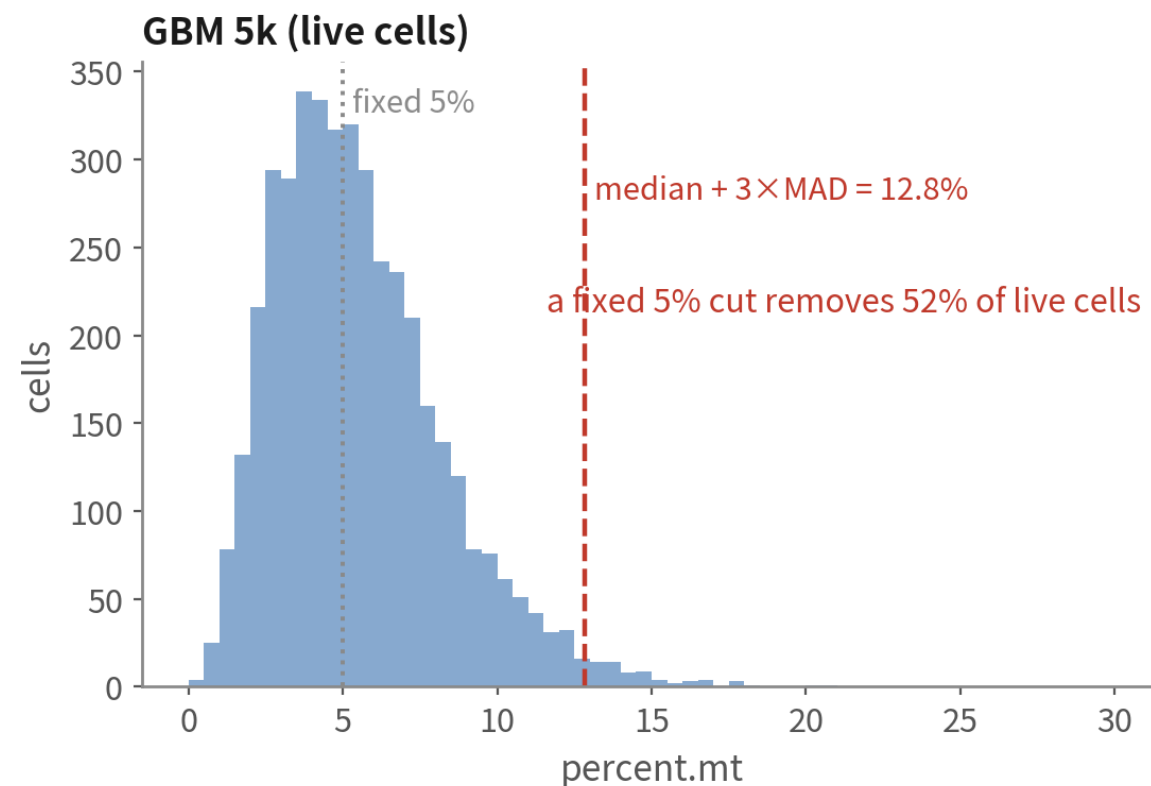
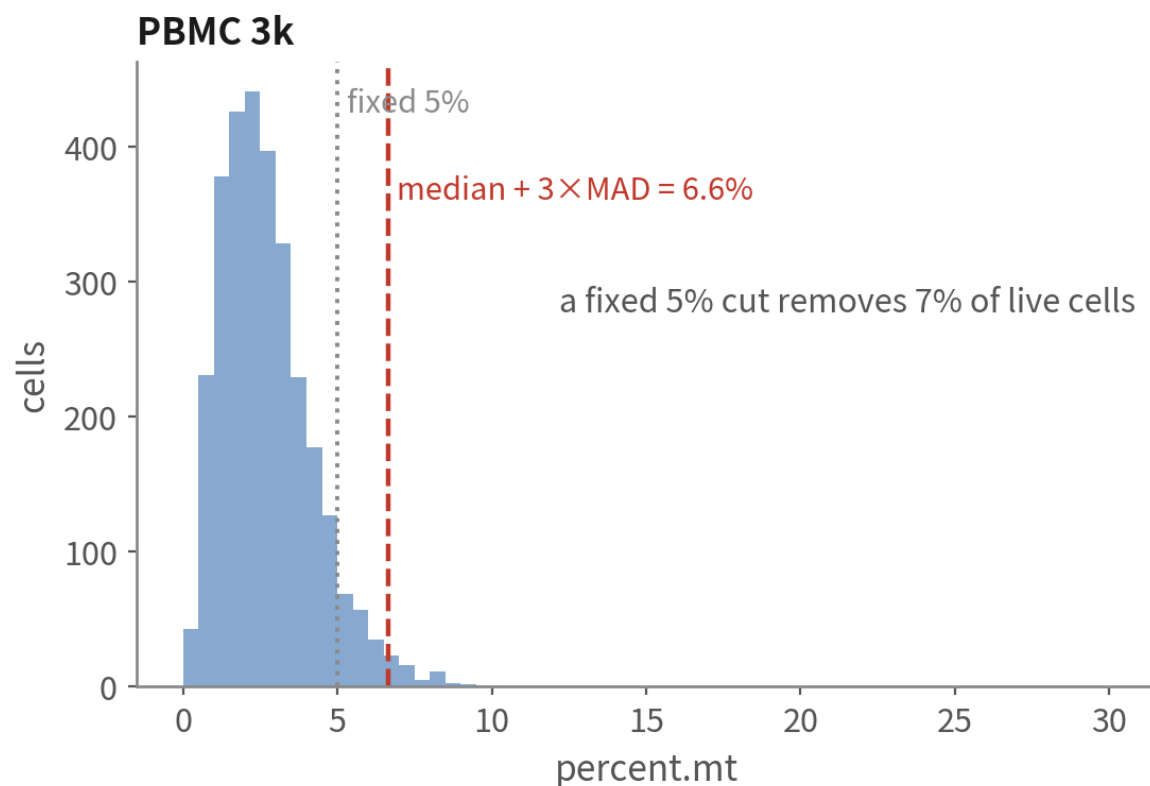
gbm                                     # 記下濾掉幾顆
```

nCount / nFeature are log-normal, so compute MAD on log10; percent.mt as is | Script 01 §3

One rule, two datasets, two different lines

Same rule, two datasets, two different lines — that is why it beats copying 5%

schematic / simulated



A copied 5% cut on GBM removes half the live cells — mostly the metabolically active malignant ones

COPY-READY METHODS SENTENCE

**"Cells were retained with $\text{percent.mt} < \text{median} + 3 \times \text{MAD}$ (= 12.8%)
and $\text{nFeature} / \text{nCount}$ within $\text{median} \pm 3 \times \text{MAD}$ on $\log_{10}$ scale;
N of 5,604 cells were kept."**

Fill N and the three numbers with your own results. Every cutoff has a number and a reason.

Doublets: flag with scDbfFinder, do not delete yet

```
library(scDbfFinder)
sce <- scDbfFinder(as.SingleCellExperiment(gbm)) # dbr 自動估
gbm$dbl.score <- sce$scDbfFinder.score
gbm$dbl.class <- sce$scDbfFinder.class
table(gbm$dbl.class) # 例: singlet ~5,000 / doublet ~300
```

singlet	doublet
5012	284

Flag and keep in metadata; decide after clustering by checking which cluster is doublets as a whole | Script 01 §4 | Deeper: B5

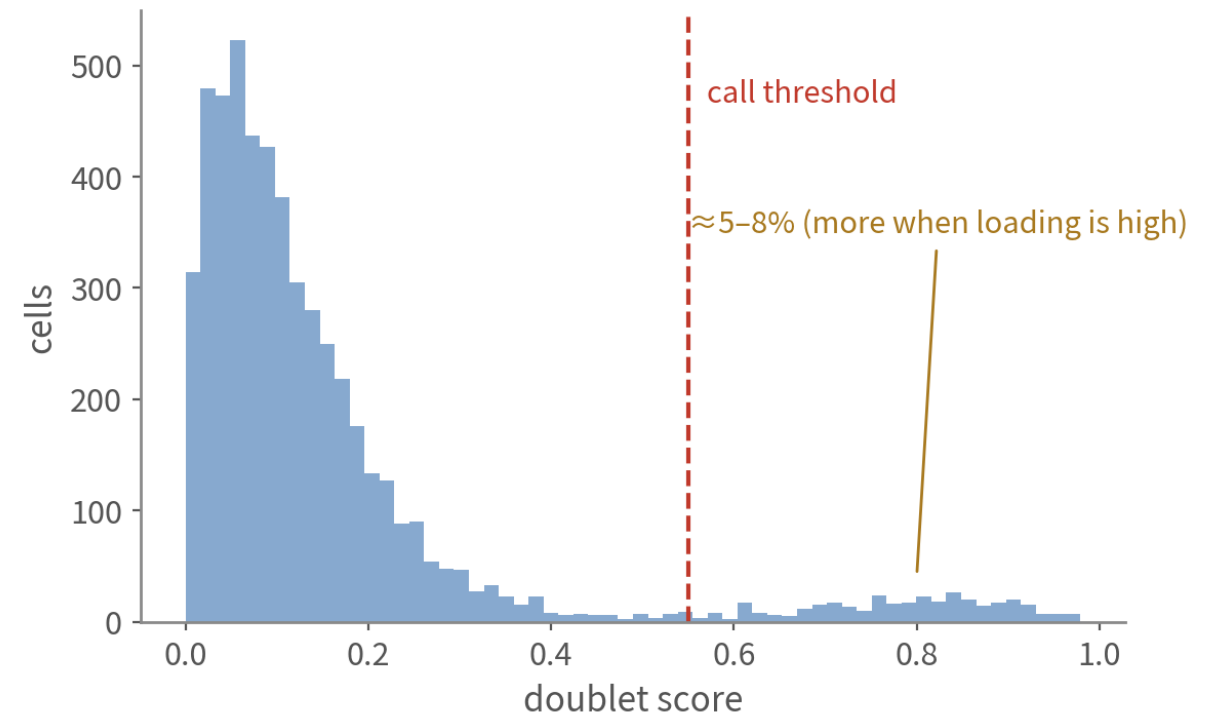
What scDblFinder does

Doublts: the junk QC metrics miss — use a dedicated tool

schematic / simulated

How scDblFinder thinks

- ① pick two cells at random, add their profiles→artificial doublets
- ② drop them into PC space and see which real cells sit next to them
- ③ real cells surrounded by artificial doublets get a high doublet score



Tumour-specific risk: malignant+macrophage doublets look like 'tumour cells expressing immune genes'. Flag first, then check per-cluster doublet rate after clustering

Tumour doublet risk is high: a malignant cell stuck to a macrophage looks like a 'tumour cell expressing immune genes'

Choosing a doublet tool, and what rate to expect

The more cells loaded, the more doublets: the 10x rule of thumb is $\approx 0.8\%$ per 1,000 cells

Tool	Idea	Strengths	Watch out
scDbtFinder (this course)	Artificial doublets + kNN classifier; expected rate via dbr	Fast, stable, Bioconductor- maintained; catches some homotypic doublets	Needs rough clusters first; run per sample in multi-sample data
DoubletFinder	Artificial doublets + pANN fraction	Seurat-native, widely cited	You supply the rate and pK; slow
Scrublet (Python)	Same idea, built into scanpy	The default in Python pipelines	Nearly blind to homotypic doublets
Hashing / demuxlet	Wet-lab: antibody tags or SNPs to split samples	The only way to catch homotypic doublets; free when pooling patients	Decided at design time not afterwards

03

Normalize, HVG, scale, PCA

Script 02: four lines of code, three sanity checks

Three problems, three steps

Problem 1: depth differs

Malignant cells carry more RNA than T cells: totals differ 5–10×. Comparing raw counts compares depth

NormalizeData

divide by total $\times 1e4$, log1p

Problem 2: too many genes

Most of 36k genes do not vary between cells—just noise

FindVariableFeatures

keep 2,000 genes varying beyond technical expectation

Problem 3: unequal scale

MALAT1 in thousands, TFs in single digits. Unscaled, PCA is hijacked by loud genes

ScaleData

z-score per gene, clip ± 10

Every later step consumes this output: PCA runs on the scale.data of the 2,000 HVGs

A malignant cell often carries ten times a T cell's RNA — step one removes exactly that difference

Four lines, one purpose each — and score the cell cycle

```
gbm <- NormalizeData(gbm)                # CP10K + log1p : 除掉深度
gbm <- FindVariableFeatures(gbm, nfeatures = 2000)  # 挑 2,000 個 HVG
head(VariableFeatures(gbm), 10)           # 先看一眼名單

gbm <- CellCycleScoring(gbm, s.features = cc.genes.updated.2019$s.genes,
                        g2m.features = cc.genes.updated.2019$g2m.genes)

gbm <- ScaleData(gbm)                     # z-score 、clip  $\pm 10$  ( 只做 HVG )
gbm <- RunPCA(gbm, npcs = 50)             # 50 個 PC 先算存著
```

Score the cycle now so you can see whether cycling cells form their own cluster; we do not regress by default | Script 02 §1 | Deeper: B6

LogNormalize or SCTransform?

Both are valid; this course uses the former because every step is visible and it plays well with pseudobulk and inferCNV

LogNormalize (this course)

CP10K + log1p; HVG via vst

- Simple, fast, interpretable; the data layer feeds FeaturePlot / DotPlot directly
- The counts layer stays untouched — pseudobulk, inferCNV and SoupX all need it
- With extreme depth differences (malignant vs T cells) a little depth effect remains — the PC1 check catches it

SCTransform

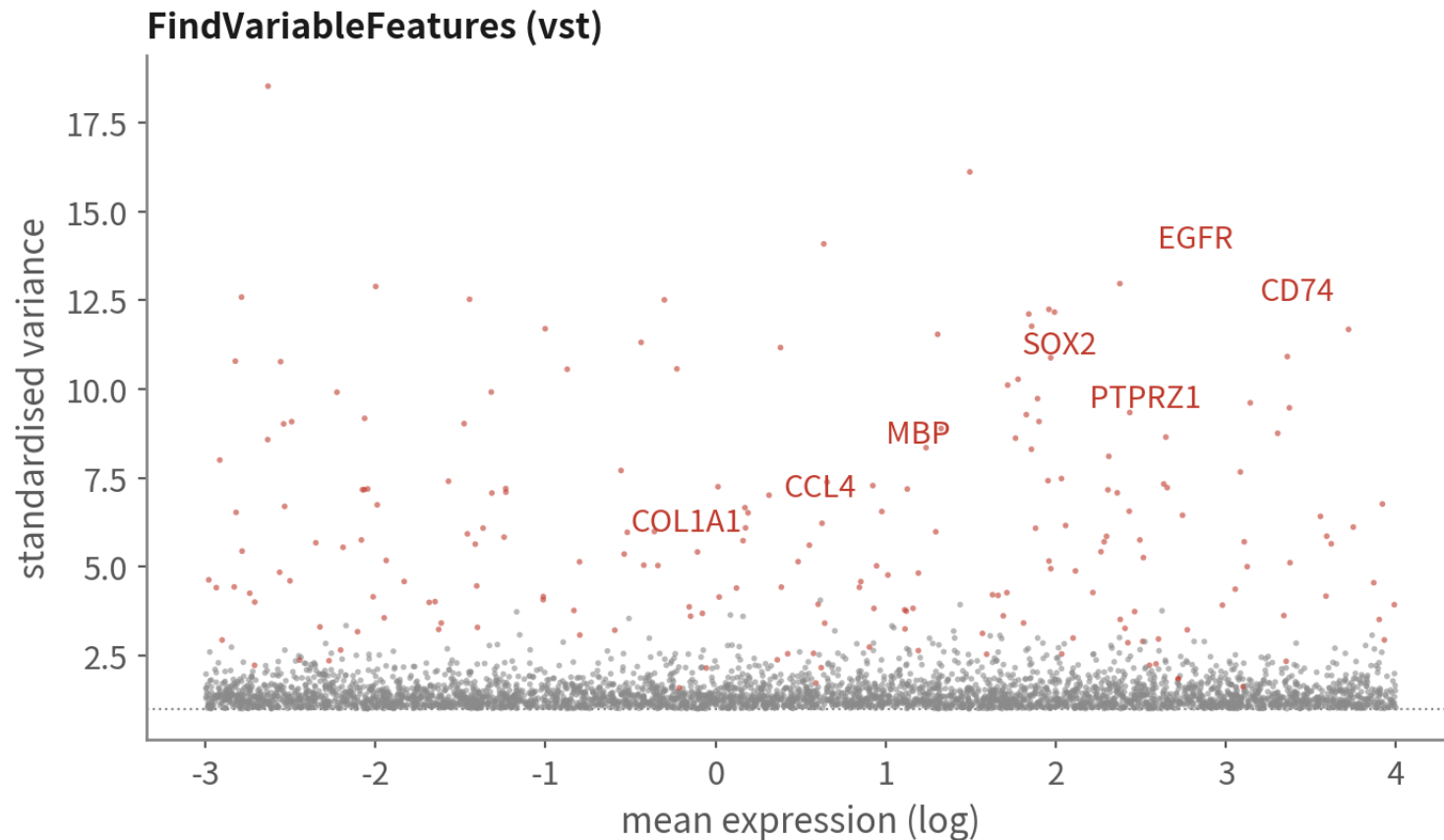
Negative-binomial regression; depth removed more cleanly

- Steadier HVGs, often cleaner clusters; v2 handles wide depth ranges well
- Adds an SCT assay; run PrepSCTFindMarkers before FindMarkers; run per sample before integration
- DE, CNV and pseudobulk still come from RNA counts — never from SCT values

The HVG list is biology in itself

HVG: 2,000 genes with something to say, out of 36,000

schematic / simulated



Who tops the GBM HVG list

- malignant-vs-normal genes (EGFR, SOX2, PTPRZ1)
- immune genes (CD74, HLA, CCL4)
- oligodendrocyte (MBP, PLP1), vascular (VWF)
- The list itself is biology—look at it before moving on

In GBM the top hits are malignant-vs-normal, immune and oligodendrocyte genes. One glance tells you who is in the data

Sanity check 1: PC1 is biology, not depth

```
print(gbm[["pca"]], dims = 1:2, nfeatures = 6)
cor(Embeddings(gbm, "pca")[, 1], gbm$nCount_RNA) # |r| < 0.3 過關
```

PC_ 1

Positive: PTPRZ1, SOX2, BCAN, OLIG1, EGFR, GPM6B ← glial / malignant

Negative: PTPRC, CD74, HLA-DRA, C1QB, AIF1, TYROBP ← immune

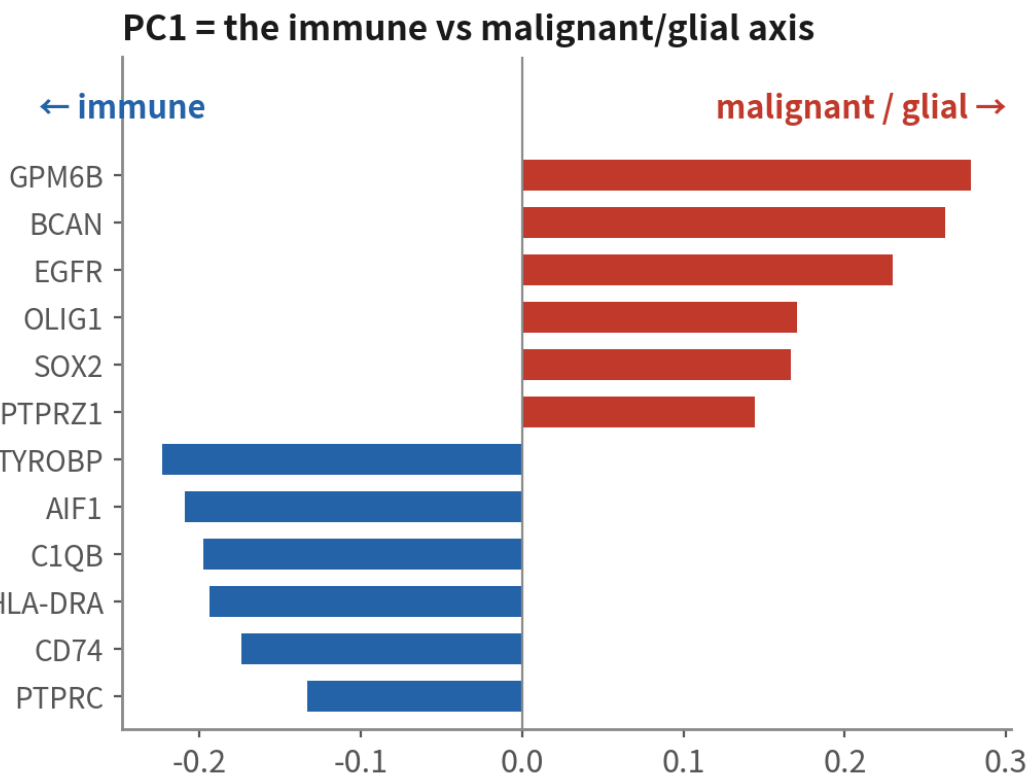
[1] 0.07

In GBM PC1 is almost always immune vs glial. Near 1 → go back and check NormalizeData | Script 02 §2 | Deeper: B7

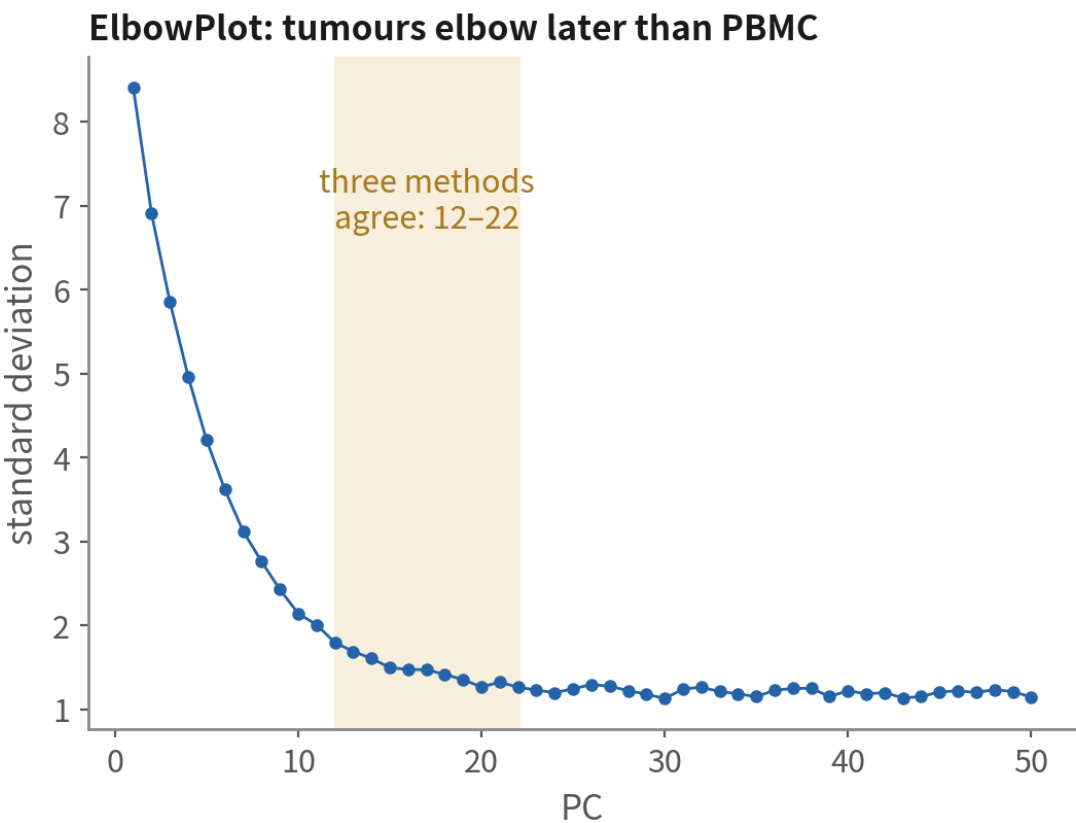
PC1 and the elbow

PCA: check PC1 is biology first, then choose how many

schematic / simulated



Tumours are structured (many types+malignant states+patient effects): use dims 20–30, more than PBMC's 10. More is safer than fewer



Tumours are structured, so the elbow comes later than PBMC. Use dims 20–30; more is safer than fewer

Sanity check 2: how many PCs

```
ElbowPlot(gbm, ndims = 50)
pct <- gbm[["pca"]@stdev / sum(gbm[["pca"]@stdev) * 100
cumu <- cumsum(pct)
which(cumu > 90 & pct < 5)[1]      # 累積 90% 且單 PC < 5%
npc <- 25                          # GBM : 三法交集約 20-30 , 取 25
```

Turn the elbow into a number you can put in the methods. Too few merges subtypes away; too many adds a little noise | Script 02 §2

Regress out the cell cycle? Check two things first

Cycling cells in tumours are biology; do not regress by default, unless the cycle hijacks the clustering

Do not regress (default)

Cycling cells form their own cluster; other clusters still split by type

- Check: `DimPlot(group.by = "Phase")` shows one small S / G2M patch
- Check: no MKI67 / TOP2A among PC1–PC5 loadings
- Handling: annotate that cluster as 'cycling X', not a new type (Trap 3)

Regress (vars.to.regress)

When every type is cut in two by the cycle

- `ScaleData(vars.to.regress = c("S.Score", "G2M.Score"))`, or regress only S – G2M to keep 'cycling vs not'
- Cost: the proliferation signal vanishes — often exactly what tumour studies want
- Same logic for `percent.mt`: regress only when it dominates a PC

04

Clustering and UMAP

Cluster in PC space; sweep resolution; QC again after clustering

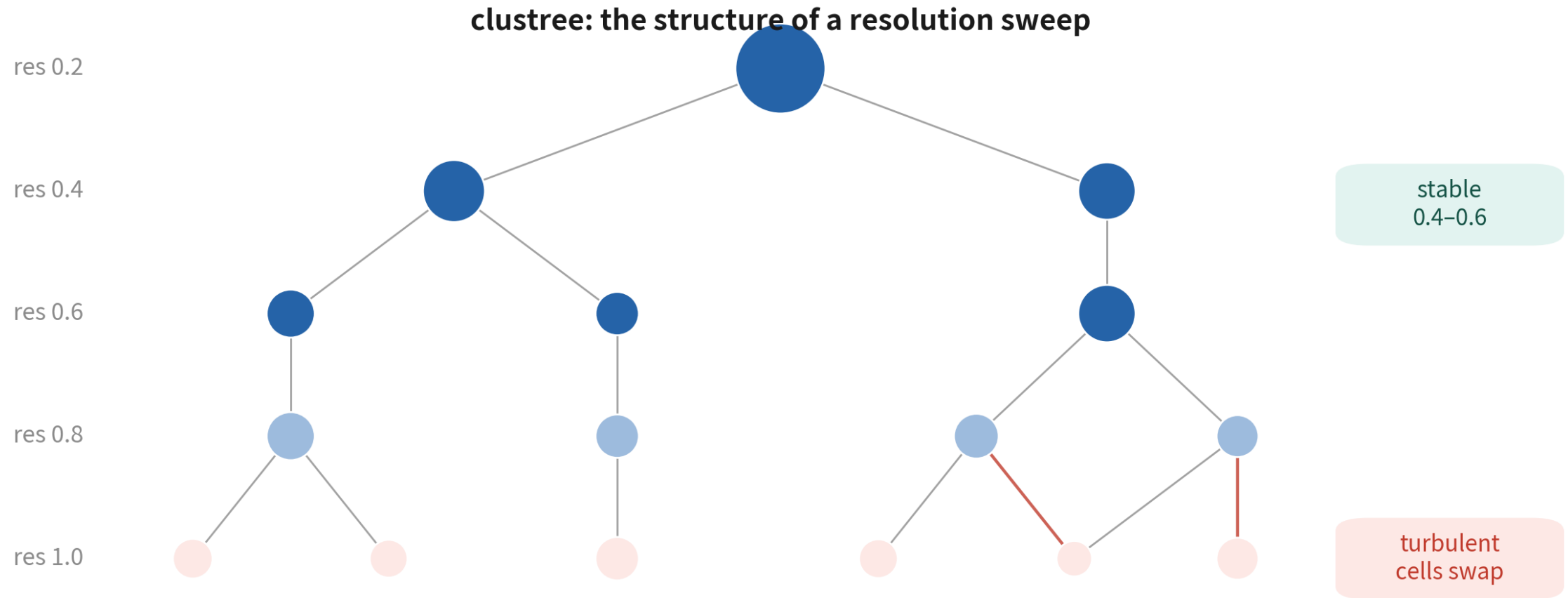
Neighbour graph → communities → projection → sweep

```
gbm <- FindNeighbors(gbm, dims = 1:npc)           # PC 空間建 KNN/SNN 圖
gbm <- FindClusters(gbm, resolution = 0.5)        # 起點，不是答案
gbm <- RunUMAP(gbm, dims = 1:npc)                # 同一組 dims

for (r in seq(0.2, 1.2, by = 0.2))
  gbm <- FindClusters(gbm, resolution = r, verbose = FALSE)
library(clustree); clustree(gbm, prefix = "RNA_snn_res.")
Idents(gbm) <- "RNA_snn_res.0.5"                 # 掃完指回定案欄位
DimPlot(gbm, label = TRUE)
```

After the sweep the active identity is the last one run — always point it back | Script 02 §3 | Deeper: B8, B9

clustree: find the stable band



Find the band where several layers agree; step back where arrows cross and cells swap. Every extra split needs marker support

Tumour data separates immune / glial / oligo at low resolution; further cuts are malignant states and immune subsets — each needs marker support

Are the clusters stable? Change the seed, change k, watch the sizes

1. 同一 resolution 換 seed: ARI 接近 1 才算穩

```
gbm <- FindClusters(gbm, resolution = 0.5, random.seed = 42, cluster.name = "seed42")  
mclust::adjustedRandIndex(gbm$RNA_snn_res.0.5, gbm$seed42) # 例: 0.9x
```

2. 換 k.param (預設 20): 群的骨架不該因此改變

```
gbm <- FindNeighbors(gbm, dims = 1:20, k.param = 30)  
gbm <- FindClusters(gbm, resolution = 0.5, cluster.name = "k30")  
table(gbm$RNA_snn_res.0.5, gbm$k30) # 對角線清楚 = 穩
```

3. 群大小: 小於 30 顆的群先存疑

```
sort(table(gbm$RNA_snn_res.0.5))
```

Ten seconds for three checks. $ARI < 0.8$, or a cluster that vanishes when k changes $\rightarrow$ an unstable cut; let markers decide whether it stays | Script 02 §3

Sanity check 3: per-cluster QC and doublets after clustering

```
gbm@meta.data |>
  group_by(cluster = RNA_snn_res.0.5) |>
  summarise(n = n(), mt = median(percent.mt), nF = median(nFeature_RNA),
            dbl = mean(dbl.class == "doublet"), cyc = mean(Phase != "G1")) |>
  arrange(desc(dbl))
```

cluster	n	mt	nF	dbl	cyc	
7	142	6.1	3120	0.62	0.31	← 60% doublets, between malignant & myeloid → drop cluster
3	410	5.4	2210	0.05	0.04	
...						

A cluster that is mostly doublets and sits between two big groups → remove and record; a cluster with runaway mt → back to QC | Script 02 §4

05

Annotation: four continents, then subtypes

Script 03: gate genes → markers → SingleR → state scores

Draw the continents with four gate genes first

FeaturePlot: draw the continents with four gate genes first

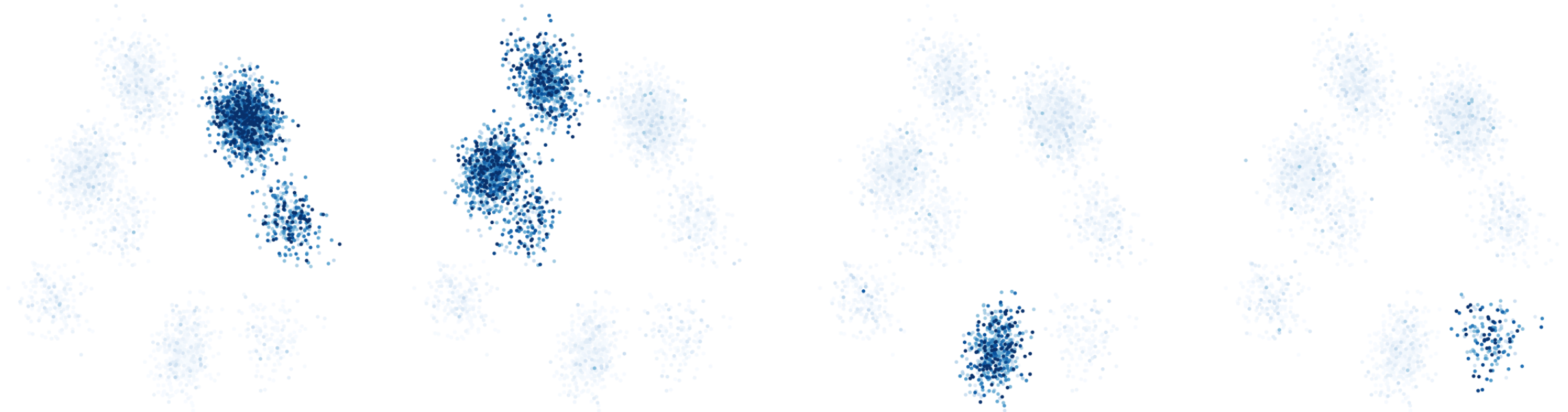
schematic / simulated

PTPRC

SOX2

MBP

PECAM1



Immune (PTPRC), glial/malignant (SOX2), oligodendrocyte (MBP),
vascular (PECAM1)—settle the four continents, then the subtypes

Immune (PTPRC), glial/malignant (SOX2), oligodendrocyte (MBP), vascular (PECAM1). Settle the four, then subtypes

Gate genes and the marker panel

```
FeaturePlot(gbm, features = c("PTPRC", "SOX2", "MBP", "PECAM1"), ncol = 4)

panel <- c("PTPRC", "CD68", "CD163", "P2RY12", "TMEM119", "CD3E",    # 免疫
          "SOX2", "OLIG2", "PTPRZ1", "EGFR",                        # 膠質 / 惡性
          "MBP", "PLP1", "PECAM1", "VWF", "PDGFRB", "RGS5",        # 寡樹突 / 血管 / 周細胞
          "MKI67", "TOP2A")                                         # 週期
DotPlot(gbm, features = panel) + RotatedAxis()
```

Blocks along the diagonal mean good clusters; PTPRC lighting three immune types is expected | Script 03 §1 | Deeper: B10

A marker panel for the GBM microenvironment

A marker panel for the GBM tumour microenvironment

population	canonical markers	one line
● immune continent	PTPRC (CD45)	gate for all immune cells
● macrophage	CD68, CD163, C1QB, AIF1	tumour-associated macrophages, the most abundant immune cells in GBM
● microglia	P2RY12, TMEM119, CX3CR1	brain-resident macrophages; distinguish from blood-derived TAM
● T cell	CD3E, CD3D, CD2 (+CD8A / IL7R)	GBM is immune-cold; T cells are usually scarce
● malignant / glial	SOX2, OLIG2, PTPRZ1, EGFR	markers only say 'glial lineage'; malignancy needs CNV (Q3)
● oligodendrocyte	MBP, PLP1, MOG	invaded normal white matter; a good inferCNV reference
● endothelial	PECAM1, VWF, CLDN5	prominent vascular proliferation in GBM
● pericyte	PDGFRB, RGS5, ACTA2	perivascular stromal cells
● cycling	MKI67, TOP2A	a state, not a type; overlays malignant cells

Microglia vs blood-derived macrophages, and cycling state vs cell type — the two pairs most often confused in tumour annotation

Witnesses per cluster: FindAllMarkers

```
markers <- FindAllMarkers(gbm, only.pos = TRUE,  
                          min.pct = 0.25, logfc.threshold = 0.5)  
markers |> group_by(cluster) |> slice_max(avg_log2FC, n = 5) |>  
  select(cluster, gene, avg_log2FC, pct.1, pct.2) |> print(n = 50)
```

cluster	gene	avg_log2FC	pct.1	pct.2	
2	C1QB	high	~0.95	~0.05	← macrophage
4	MBP	high	~0.98	~0.08	← oligodendrocyte
0	PTPRZ1	high	~0.9	~0.2	← glial (malignant pending CNV)

Read in order: avg_log2FC → pct.1 / pct.2 → p. pct.2 near 0 is the clean marker | Script 03 §2

Filtering the marker table: four columns, four questions

Look at p last — with 5,000 cells nearly everything is significant

Column	It answers	A good marker	Trap
avg_log2FC	How much higher than the rest	> 1 (over two-fold)	Low genes get their FC squashed by the pseudocount
pct.1	Fraction expressing inside the cluster	> 0.5	A 'marker' at pct.1 = 0.3 describes part of the cluster — a hidden subset or doublets
pct.2	Fraction expressing elsewhere	< 0.1	Ambient RNA inflates pct.2; MBP at pct.2 = 0.3 does not mean oligodendrocytes everywhere
p_val_adj	Whether the difference is chance	< 0.05 (almost all are)	Cells in a cluster are not independent; p is overstated by orders of magnitude — use only as a sieve

Second witness: SingleR

```
library(SingleR); library(cellidex)
ref <- cellidex::HumanPrimaryCellAtlasData()
pred <- SingleR(test = GetAssayData(gbm, layer = "data"), ref = ref,
               labels = ref$label.main, clusters = gbm$RNA_snn_res.0.5)
data.frame(cluster = rownames(pred), SingleR = pred$labels)
```

```
cluster SingleR
0      Astrocyte      ← no "GBM malignant" in the reference – forced into the nearest normal type
2      Macrophage
4      Oligodendrocytes
5      Endothelial_cells
```

Accurate for immune, oligo and vascular; malignant cells come out as astrocyte or neural progenitor — a reminder, not an answer | Script 03 §3

Where automatic annotation stands (tumour data as the example)

Automatic annotation (SingleR /Azimuth) on tumour data

- ✓ Immune cells: accurate—references (HPCA, Blueprint) cover them well
- ✓ Oligodendrocytes, endothelium: accurate
- ✗ Malignant cells: no 'GBM malignant' in any reference—forced onto the nearest normal type (astrocyte, neural progenitor)
- Verdict: use it as a second witness; call malignancy separately

Cross-check table (example)

cl.	manual	SingleR	verdict
2	macrophage	Macrophage	✓
3	T cell	T_cells	✓
4	oligo	Oligodendrocyte	✓
0	malig. AC-like	Astrocyte	CNV decides
1	malig. OPC-like	Neural progenitor	CNV decides
5	endothelial	Endothelial	✓

Three-column cross-check: agreeing clusters pass; glial clusters wait for CNV (Q3)

Annotation strategies: four roads, usually walked together

No single road is reliable alone; this course goes gate → markers → reference → scores

Strategy	How	Strength	Weakness	Tools
Manual markers	Dotplot / FeaturePlot per cluster against literature markers	Interpretable; finds novel types	Slow, subjective, cluster-dependent	Seurat
Reference classification	Correlate each cell / cluster with an annotated reference	Fast, reproducible; accurate for normal types	Types absent from the reference (tumour) get forced into the nearest	SingleR, scType, CellTypist
Gene-set scoring	Mean / AUC of a gene set as a score	Continuous, comparable suits states	High score ≠ type; thresholds are subjective	AddModuleScore, UCell, AUCell
Atlas mapping	Project cells onto a large atlas and inherit labels	Finest subsets, consistent labels	Needs a good atlas malignant cells cannot be mapped	Azimuth, scArches

Choosing an automatic annotation tool

Run three on the same GBM data: normal types agree > 90%; the disagreements are where a human must look

Tool	Idea	Reference	Unit	Best for / watch out
SingleR (this course)	Spearman correlation + fine-tuning	cellDex (HPCA, Blueprint, Monaco immune)	Cell or cluster	Default for normal types; an immune reference splits T subsets
scType	Marker database + cluster-level scoring	Built-in human / mouse tissue markers	Cluster	Results in seconds; brain table has no tumour
CellTypist	Logistic-regression models	Immune atlas, multiple human tissues	Cell	Strongest on immune subsets (Python)
Azimuth	Anchor mapping onto a reference UMAP	PBMC, brain, lung kidney, marrow...	Cell	Comes with UMAP coordinates; the brain reference does not apply to GBM malignant cells
scANVI / scArches	Semi-supervised deep model	Any atlas you train	Cell	Cross-technology, large scale; needs GPU

Assign names: provisional and honest

```
new.ids <- c("0" = "Glial (CNV pending)", "1" = "Glial (CNV pending)",  
            "2" = "Macrophage", "3" = "Microglia", "4" = "Oligodendrocyte",  
            "5" = "Endothelial", "6" = "Pericyte", "8" = "T cell")  
gbm <- subset(gbm, idents = "7", invert = TRUE) # 移除 doublet 群  
gbm <- RenameIdents(gbm, new.ids)  
gbm$celltype <- Idents(gbm) # 存進 metadata  
DimPlot(gbm, label = TRUE, repel = TRUE) + NoLegend()
```

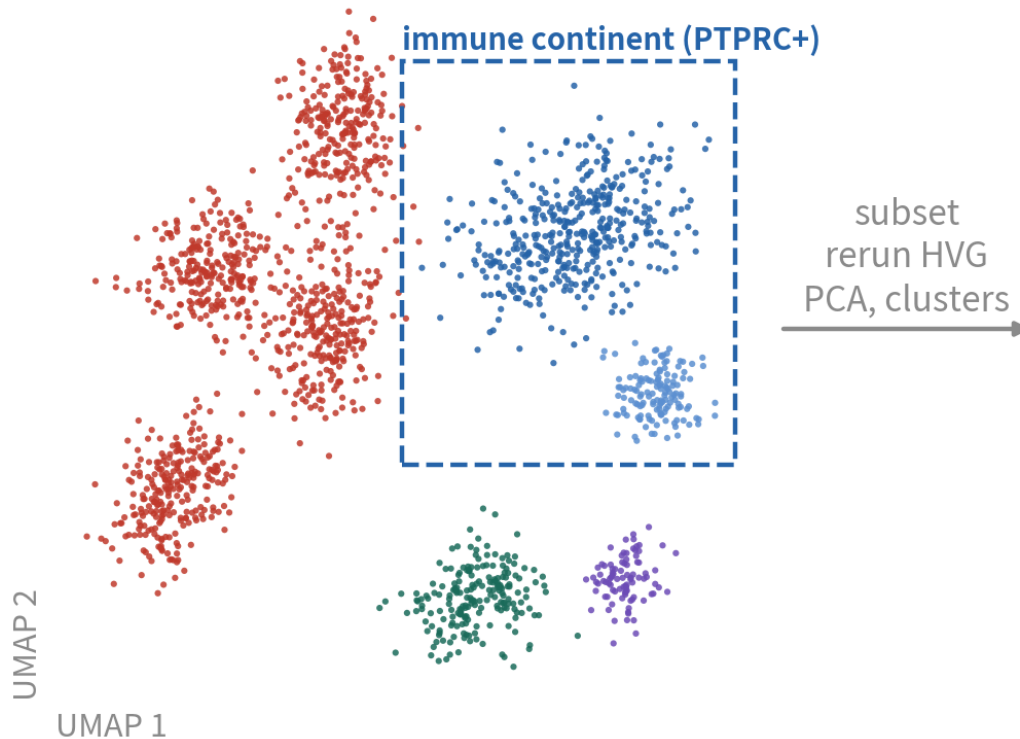
Match to your own DotPlot before assigning; number-to-name maps differ for every dataset | Script 03 §4

Hierarchical annotation: continents settled, now the subsets

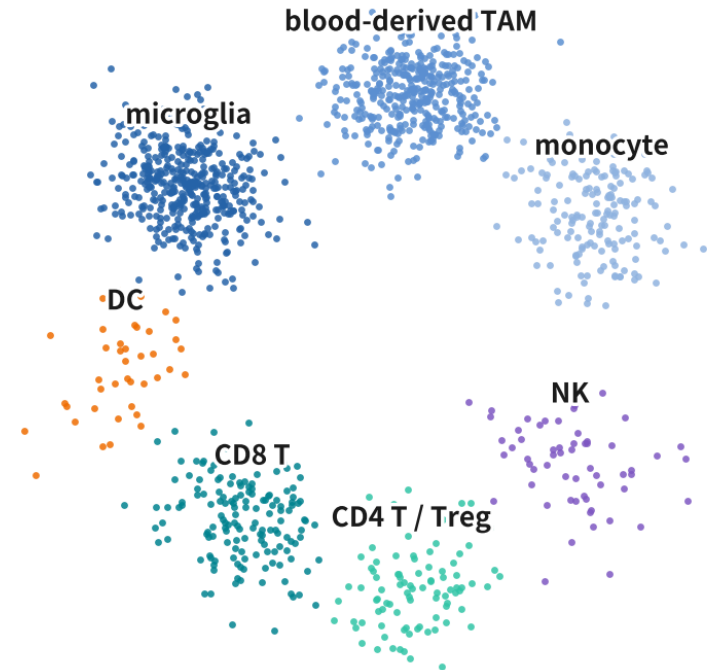
Hierarchical annotation: continents first, subsets next, re-embed at every level

schematic / simulated

Level 1: four continents



Level 2: immune subsets (recomputed on immune cells only)



Why recompute

- 1 Global HVGs are dominated by malignant-vs-immune; within-immune differences never make the top 2,000 same for PC subset signal sits beyond PC 20
- 2
- 3 Rerun on immune cells alone and CD4/CD8, microglia vs blood-derived TAM separate

Take the immune continent out and rerun HVG, PCA and clustering — only then do CD4 / CD8 and microglia / blood-derived macrophages separate. Malignant 'subsets' use state scores, not clusters

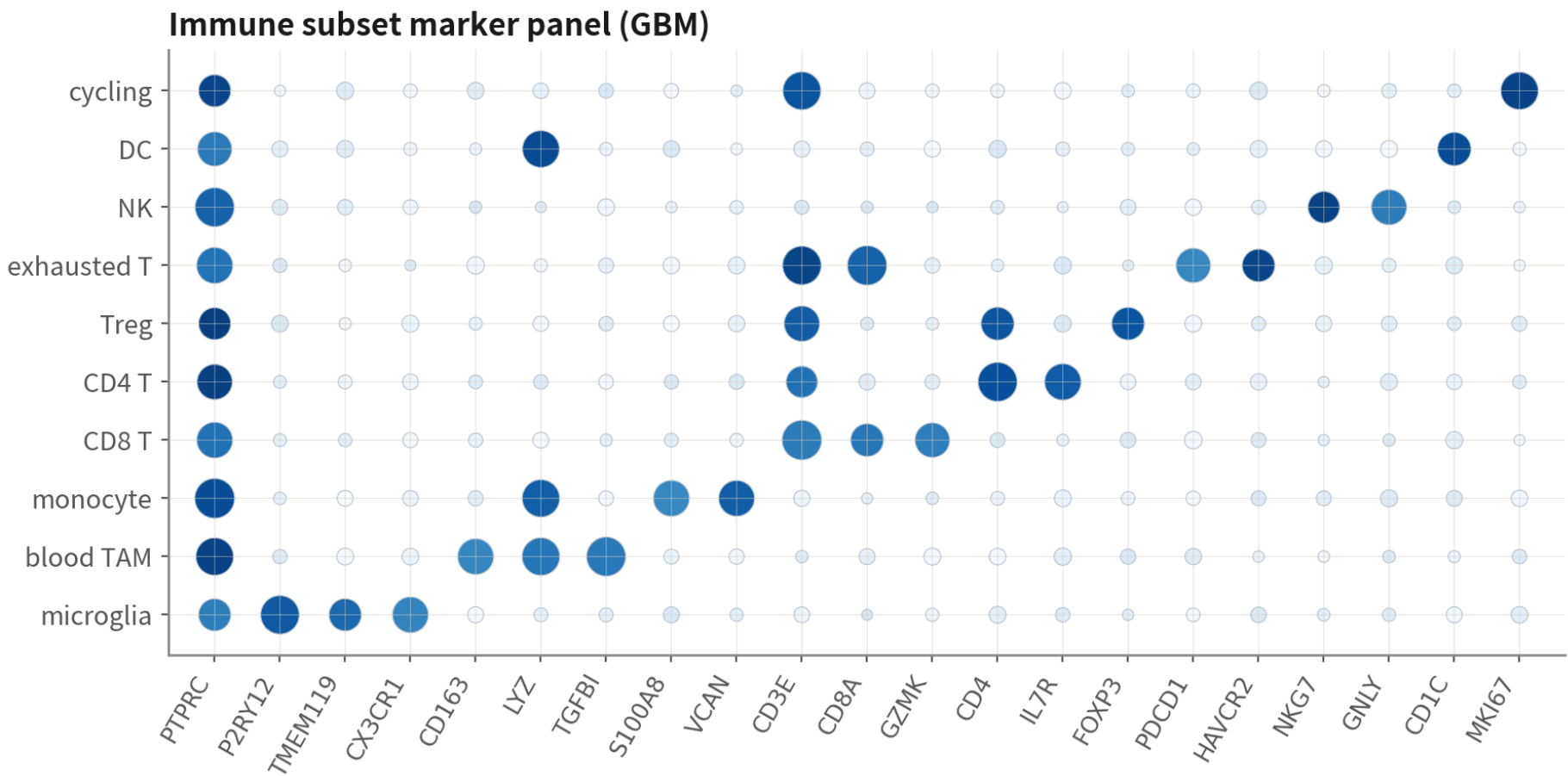
Subsets: subset, then rerun the whole pipeline

```
imm <- subset(gbm, celltype %in% c("Macrophage", "Microglia", "T cell"))
imm <- NormalizeData(imm) |> FindVariableFeatures(nfeatures = 2000) |>
  ScaleData() |> RunPCA(npcs = 30)
ElbowPlot(imm) # 亞群通常 15-20 個 PC 就夠
imm <- FindNeighbors(imm, dims = 1:20) |> FindClusters(resolution = 0.6) |>
  RunUMAP(dims = 1:20)
imm.panel <- c("P2RY12", "TMEM119", "CX3CR1", # 小膠質
  "CD163", "LYZ", "TGFB1", "S100A8", "VCAN", # 血液來源 TAM / 單核球
  "CD3E", "CD8A", "GZMK", "CD4", "IL7R", "FOXP3",
  "PDCD1", "HAVCR2", "NKG7", "GNLY", "CD1C", "MKI67")
DotPlot(imm, features = imm.panel) + RotatedAxis()
# 只想切某一群、不重跑 : FindSubCluster(gbm, cluster = "2", graph.name = "RNA_snn")
```

Always recompute HVGs and PCA after subsetting; circling on the old UMAP is a common error. Do not split subsets under ~200 cells | Script 03 §4b

The immune subset panel: the diagonal is the evidence

Subset annotation: the dotplot diagonal is the evidence



schematic / simulated

How to read

PTPRC lights the whole column: a gate, not a subset marker
Microglia vs blood TAM: P2RY12/TMEM119 vs CD163/TGFBI
Exhausted T=CD8 +PDCD1/HAVCR2, not a new type
Cycling immune lights only MKI67: a state—merge back

Microglia vs blood-derived TAM, CD8 vs exhausted CD8, Treg — every cut needs its own markers; a group lit only by MKI67 is a state, merge it back

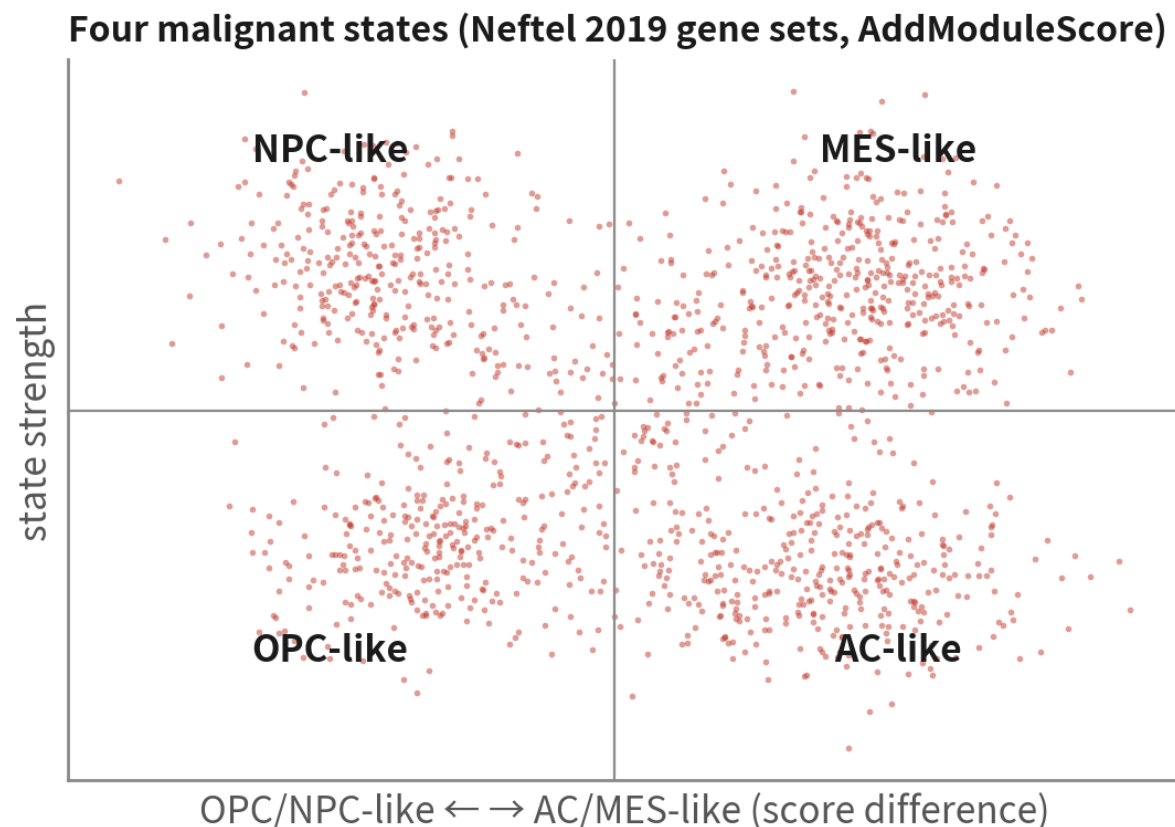
Malignant state scores (Neftel 2019)

```
neftel <- list(                                     # 每組取前 8 個示範
  MES = c("CHI3L1", "ANXA2", "ANXA1", "CD44", "VIM", "MT2A"),
  AC  = c("CST3", "S100B", "SLC1A3", "HOPX", "MT3", "SPARCL1"),
  OPC = c("BCAN", "PLP1", "GPR17", "FIBIN", "LHFPL3", "OLIG1"),
  NPC = c("DLL3", "DLL1", "SOX4", "TUBB3", "HES6", "TAGLN3"))
glial <- subset(gbm, idents = "Glial (CNV pending)")
glial <- AddModuleScore(glial, features = neftel, name = names(neftel))
FeaturePlot(glial, features = paste0(names(neftel), 1:4), ncol = 4)
```

Use the paper's supplementary table for full sets (50 genes each); eight per set here for illustration | Script 03 §5 | Deeper: B13

Why score states rather than cluster again

schematic / simulated



Why score states instead of clustering again

- Malignant states are continuous; hard clusters invent boundaries
- Scoring with published gene sets makes results comparable across studies
- One patient's tumour holds all four states at once—one source of therapy resistance
- Score cell cycle (G2M/S) too, or cycling cells form their own cluster

Malignant states are continuous; scores keep that continuity and are comparable across studies. Read the cycle score alongside

Three ways to score: AddModuleScore, UCell, AUCell

```
# 1. AddModuleScore ( Seurat ) : 基因集平均 - 表現量相近的對照基因平均。快，但受深度影響  
glial <- AddModuleScore(glial, features = nefte1, name = names(nefte1))
```

```
# 2. UCell : 以基因在每顆細胞內的「排名」算 Mann-Whitney U 分數，對深度與稀疏穩健  
library(UCell)  
glial <- AddModuleScore_UCell(glial, features = nefte1, name = "_UCell")
```

```
# 3. AUCell : 每顆細胞的基因排名前 5% 裡有多少基因集成員 ( AUC ) ; SCENIC 也用它  
library(AUCell)  
rank <- AUCell_buildRankings(LayerData(glial, layer = "counts"))  
auc <- AUCell_calcAUC(nefte1, rank)
```

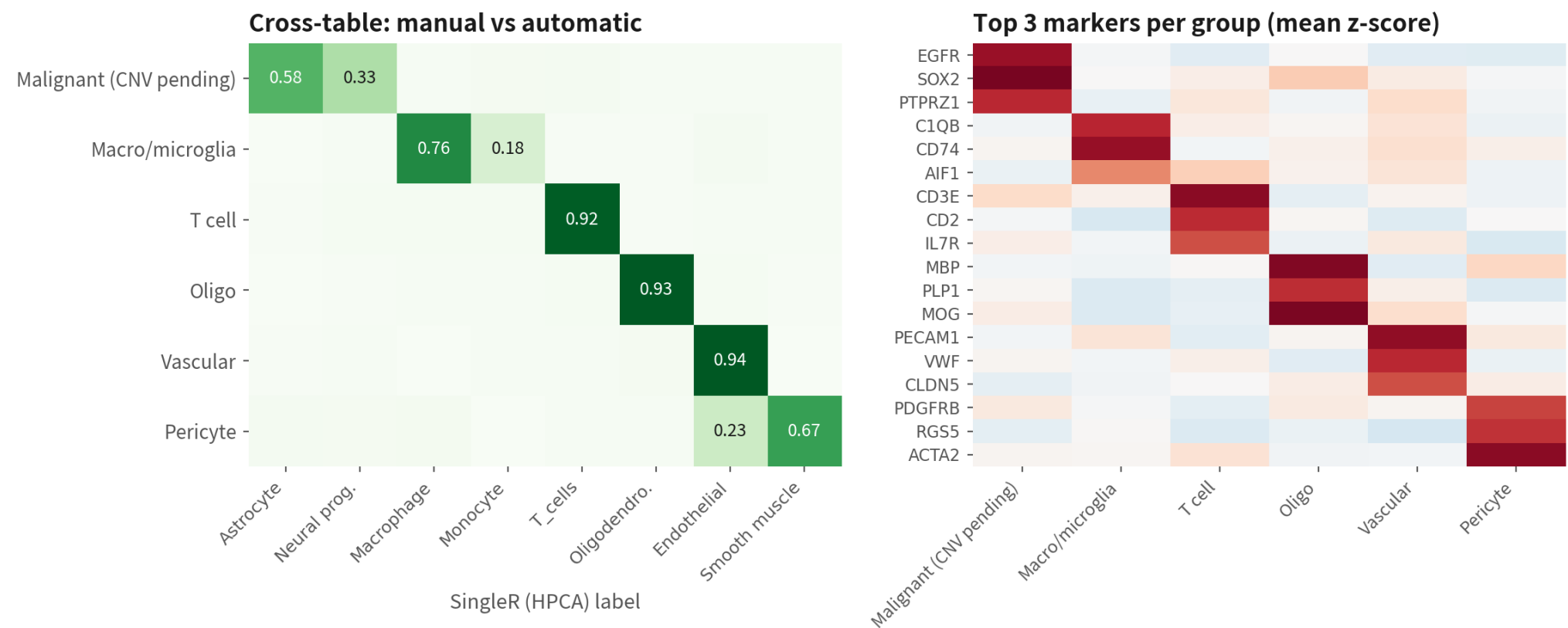
```
FeaturePlot(glial, features = paste0(names(nefte1), "_UCell"), ncol = 4)  
cor(glial$MES1, glial$MES1_UCell) # 三種算法的排序通常高度一致
```

Report the algorithm and the gene-set source; when the three disagree, trust UCell (least depth-sensitive) | Script 03 §5

Annotation quality control: the cross-table and the marker heatmap

Two pieces of evidence after annotation: the cross-table and the marker heatmap

schematic / simulated



Left: a clean diagonal; malignant splitting into astrocyte/neural progenitor is expected (no tumour in the reference). Right: blocks along the diagonal, no group lighting everything

Left: a clean manual-vs-automatic diagonal with every deviation explained; right: top markers light in blocks, no group lights everything. Both go into the supplement

QC code: cross-table, heatmap, cluster tree, silhouette

```
# 交叉表：手動標籤 vs SingleR (每列加總為 1)
round(prop.table(table(gbm$celltype, gbm$singler), 1), 2)

# 每群 top 5 marker 熱圖 (每群最多抽 100 顆，否則圖太重)
top5 <- markers |> group_by(cluster) |> slice_max(avg_log2FC, n = 5)
DoHeatmap(subset(gbm, downsample = 100), features = top5$gene, group.by = "celltype")

# 分群樹：型別之間的相似結構是否合理 (免疫一支、膠質一支)
gbm <- BuildClusterTree(gbm, dims = 1:20); PlotClusterTree(gbm)

# 輪廓係數：每群在 PC 空間裡有多「自成一格」 (< 0.1 的群要存疑)
library(cluster)
sil <- silhouette(as.integer(Ids(gbm)), dist(Embeddings(gbm, "pca")[, 1:20]))
tapply(sil[, "sil_width"], Ids(gbm), mean)
```

Four checks, four questions: does the automatic label agree, are markers clean, is the structure sensible, are clusters separable | Script 03 §5b

Naming convention: keep three label levels in metadata

Papers use level 2, supplementary tables level 3, statistical tests often only level 1

Column	Content	Example	Rule
celltype_l1	Continent	Immune / Malignant / Oligo / Vascular / Stromal	Five to seven; composition analysis uses this
celltype_l2	Type	Microglia, blood-derived TAM CD8 T, malignant (CNV)	Each backed by ≥ 2 markers; names aligned to Cell Ontology
celltype_l3	State / subset	Exhausted CD8 T, cycling TAM, MES-like	States as 'adjective + type'; do not coin new nouns
celltype_conf	Confidence	high / medium / unresolved	Write unresolved when unsure; never force

Common mis-annotations: five 'novel types' and what they really are

Looks like	Actually	How to catch it	What to do
Tumour cells expressing immune genes	Malignant + macrophage doublets	High per-cluster doublet rate, high nCount, sits between two big groups	Remove the cluster and record it
A stem-like subset	A cycling existing type	Markers are only MKI67 / TOP2A; Phase nearly all S/G2M	Merge back, label 'cycling'
A stress-response subset	Dissociation artefact	FOS / JUN / HSPA1A high; a slice inside every type	Regress or annotate; not biology
Undifferentiated / unknown type	Low-quality cells	Lowest nFeature, highest mt, no marker lights	Back to QC
Immune cells expressing oligodendrocyte genes	Ambient RNA	MBP / PLP1 a little in every group high pct.2	Rerun after SoupX

Deliverables: the three plots and one table of a Figure 1

```
p1 <- DimPlot(gbm, group.by = "celltype", label = TRUE,
              repel = TRUE) + NoLegend()
p2 <- DotPlot(gbm, features = panel, group.by = "celltype") + RotatedAxis()
p3 <- VlnPlot(gbm, features = c("nFeature_RNA", "percent.mt"),
              group.by = "celltype", pt.size = 0, ncol = 2)
(p1 | p2) / p3 # patchwork 排版
ggsave("output/figs/03_fig1_annotation.pdf", width = 14, height = 9, bg = "white")

comp <- gbm@meta.data |> dplyr::count(celltype) |>
  mutate(pct = round(100 * n / sum(n), 1))
write.csv(comp, "output/03_composition.csv", row.names = FALSE)
```

celltype	n	pct
Malignant (glial, CNV pending)	3xxx	6x.x
Macrophage / microglia	1xxx	2x.x
Oligodendrocyte	3xx	x.x
...		

UMAP for position, dotplot for evidence, violin for QC consistency, the table for numbers. Hand in all four and half the first-round questions disappear | Script 03 §6

Wrap up: save the object, record versions

```
saveRDS(gbm, "output/gbm_annotated.rds")  
write.csv(markers, "output/03_markers.csv", row.names = FALSE)  
sessionInfo() # 收進附錄；審稿人的好朋友
```

Every file in output/ can be regenerated from the scripts in R/ — that is what reproducible means | Script 03 §6

06

Where people get hurt

The three most common hands-on traps in tumour data



Trap 1: copying the 5% mito cut

What it is

Applying the PBMC tutorial's 5% to GBM.



Why it happens

Every tutorial uses PBMC; 5% looks like common sense.

What it costs you

Half the live cells gone, mostly active malignant ones. Immune proportions get inflated and every downstream composition analysis is skewed.

Trap 2: a doublet cluster reported as 'immune-like tumour cells'

What it is

A cluster expressing both SOX2 and CD74 written up as an 'antigen-presenting malignant subset'.



Why it happens

The markers look exciting and there is literature to cite.

What it costs you

Sixty percent of the cluster is scDbtFinder-flagged and it sits between malignant and myeloid. Sanity check 3 exists for exactly this.

Trap 3: cycling cells reported as a new type

What it is

An MKI67/TOP2A-high island named a 'stem-like subpopulation'.



Why it happens

Cell-cycle signal is strong and easily dominates clustering.

What it costs you

It is a state, not a type: dividing malignant cells and macrophages mixed together. Check the Phase column first, then decide whether to regress.

Q2 checklist

Q2 checklist: all nine boxes before the object goes to Q3



raw object saved
before filtering



each QC cutoff has a
number and a reason (MAD)



doublet scores
stored in metadata



$|\text{cor}(\text{PC1}, \text{nCount})| < 0.3$



dims justified (elbow
+quantified cutoff)



resolution swept; stable
band backed by markers



four continents settled
with gate genes



every name has two witnesses
(markers+automatic)



malignant label provisional:
'glial/pending CNV'

All nine boxes before the object goes to Q3

ONE-LINE SUMMARY

Every parameter is an assumption: a number, a reason, a record.

And label malignancy honestly as 'pending CNV'.

Do that and your pipeline is reproducible and defensible. Q3 fills in the 'pending CNV'.

Check yourself 1

GBM: median percent.mt 7%, median + 3×MAD = 13%. Where does the cutoff go?

- A The conventional 5%
- B 13%, with the number and reason in the methods
- C Skip the percent.mt filter

Answer B. A threshold is a property of the data. 5% removes most live cells; no filter keeps the dying-cell mode.

Check yourself 2

A cluster expresses both SOX2 and C1QB; scDbtFinder flags 65% of it. What now?

- A An exciting new subset — report it
- B Remove the whole cluster and record why
- C Remove only the flagged 65%

Answer B. Mostly flagged and sitting between two big groups — the whole cluster is a doublet product; the remaining 35% are the ones the tool missed, not real cells.

Check yourself 3

SingleR labels cluster 0 as Astrocyte; markers are SOX2 / PTPRZ1 / EGFR. The honest label?

A Astrocyte

B Malignant

C Glial (malignancy pending CNV)

Answer C. No GBM malignant cells exist in the reference, so SingleR picks the nearest; markers only give lineage. Malignancy needs CNV evidence — Q3.

NEXT

Q3: Hands-on II — multi-sample analysis

Switching to four patients, core vs periphery (GSE84465). inferCNV fills in 'pending CNV', pseudobulk makes DE patient-level, and you get a downstream road map.